LAB EXPERIMENT

16. Implement a Sobel algorithm using Open CV to filter the input image.

Code:

```
import cv2

img = cv2.imread(r"C:\Users\Admin\Downloads\images.jpg")

if img is None:
    print("Error: Image not found!")
else:
    # Convert image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Apply Sobel filter in X and Y directions
    sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
    sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

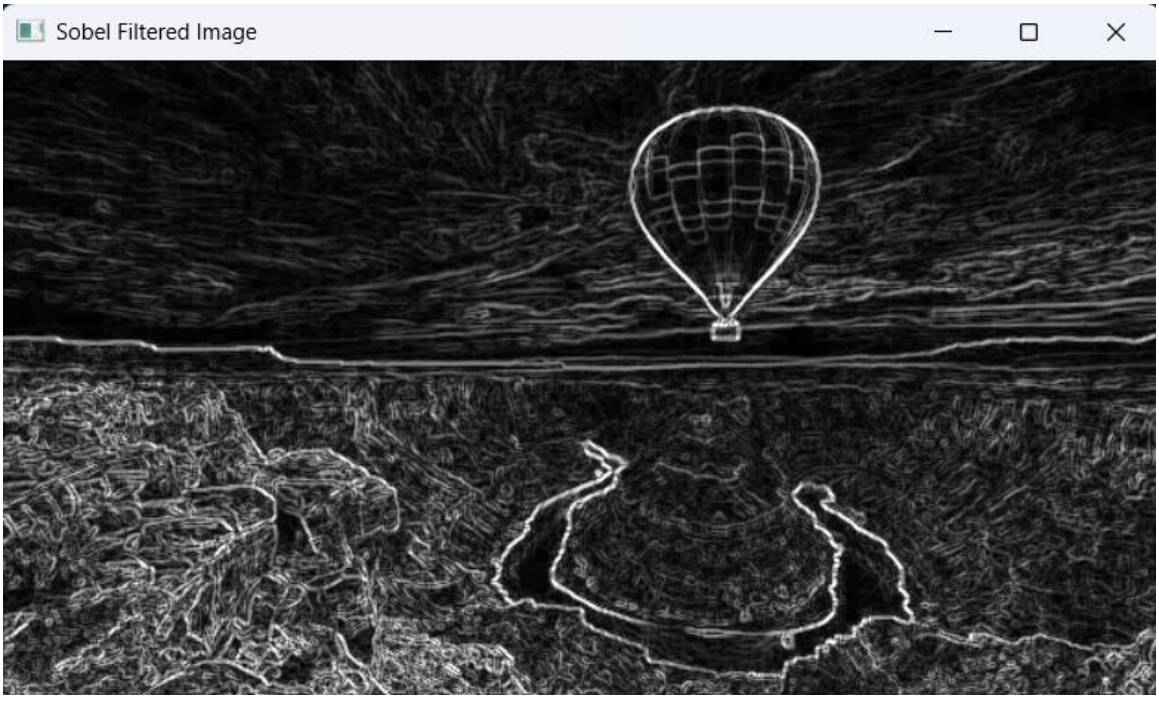
    # Convert results to displayable images
    sobel_x = cv2.convertScaleAbs(sobel_x)
    sobel_y = cv2.convertScaleAbs(sobel_y)

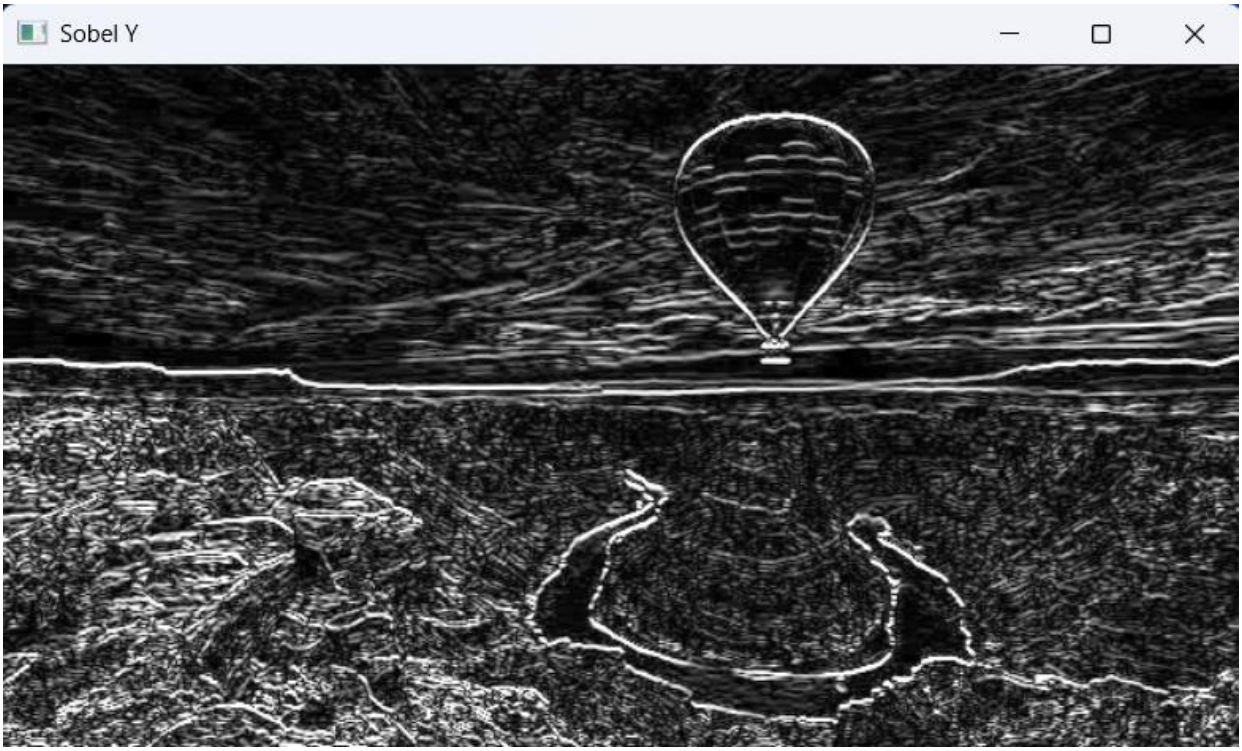
    # Combine both directions
    sobel = cv2.addWeighted(sobel_x, 0.5, sobel_y, 0.5, 0)

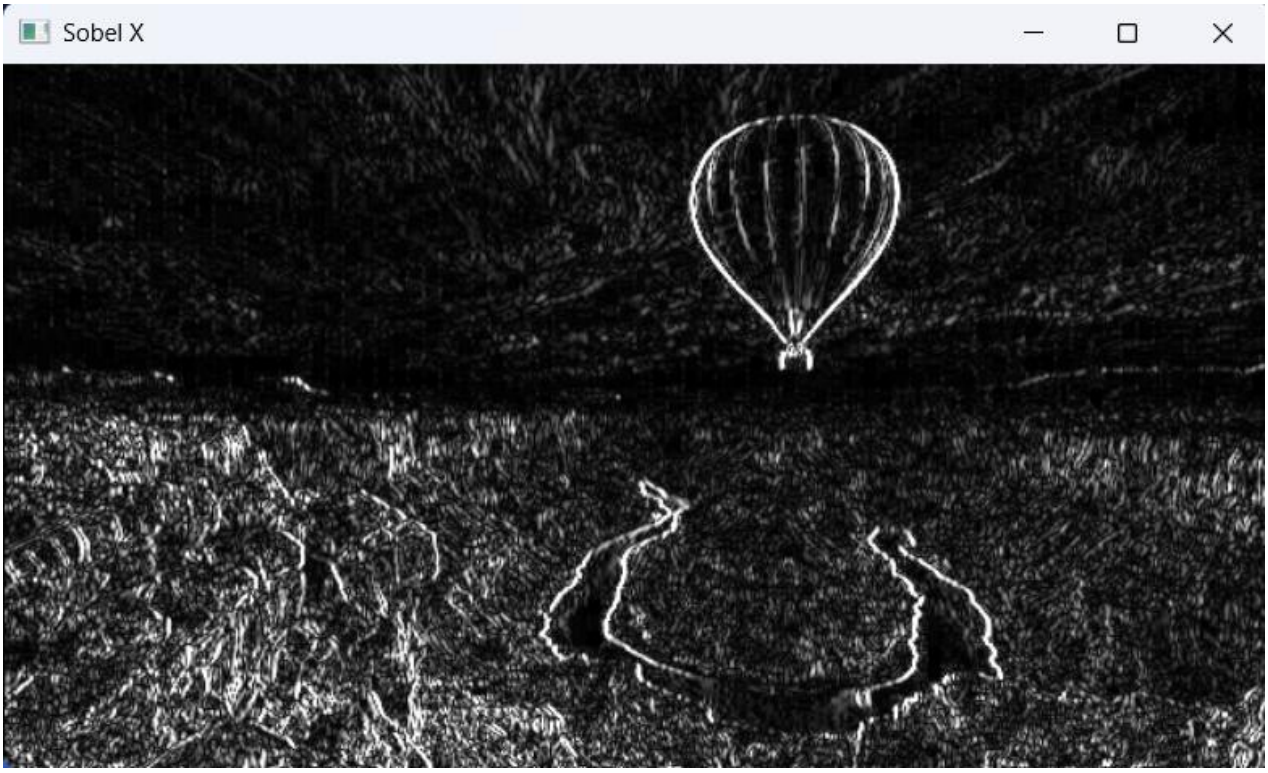
    # Display images
    cv2.imshow("Original Image", img)
    cv2.imshow("Sobel X", sobel_x)
    cv2.imshow("Sobel Y", sobel_y)
    cv2.imshow("Sobel Filtered Image", sobel)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```


Original Image


Sobel Filtered Image


Sobel Y


Sobel X